

# RAG Fresher Interview Questions & Answers

1. What is RAG?

Answer: RAG (Retrieval-Augmented Generation) ek technique hai jisme LLM pehle relevant documents retrieve karta hai aur phir un documents ki help se answer generate karta hai.

2. Why do we need RAG?

- Hallucinations kam hoti hain.
- Latest data use kar sakte hain.
- LLM ko retrain/fine-tune karne ki zarurat nahi padti.

3. Explain the RAG pipeline.

1. Documents load karo.
2. Documents ko chunks mein divide karo.
3. Chunks ke embeddings banao.
4. Embeddings ko vector database mein store karo.
5. User query ka embedding banao.
6. Similar chunks retrieve karo.
7. Retrieved context + query ko LLM ko bhejo.
8. LLM final answer generate karta hai.

4. What is an embedding?

Text ko numbers (vectors) mein convert karna, taaki semantic similarity compare ki ja sake.

5. What is a vector database?

Embeddings ko efficiently store aur search karne ke liye database.

Examples: FAISS, ChromaDB, Pinecone.

6. What is chunking?

Bade documents ko chhote pieces (chunks) mein todna.

7. Why is chunking important?

LLMs ki context limit hoti hai aur chhote chunks se retrieval quality improve hoti hai.

8. What is cosine similarity?

Do vectors kitne similar hain, ye measure karta hai.

9. Difference between SQL DB and Vector DB?

SQL DB → Structured data (tables).

Vector DB → Embeddings aur similarity search.

10. What is semantic search?

Keyword match ki jagah meaning ke basis par search karna.

11. Difference between keyword search and semantic search?

Keyword search exact words dekhta hai.

Semantic search meaning samajhta hai.

12. Which embedding model have you used?

Examples: text-embedding-3-small, all-MiniLM-L6-v2, BAAI/bge-small-en.

13. Which vector database have you used?

Honestly answer based on your project, e.g., FAISS or ChromaDB.

14. What libraries have you used?

LangChain, Hugging Face, FAISS, ChromaDB, OpenAI APIs.

15. How does similarity search work?

User query ka embedding banta hai, phir vector DB sabse similar document embeddings retrieve karta hai.

16. What is LangChain?

LLM applications banane ka framework jo prompts, retrievers, vector stores aur chains ko connect karta hai.

17. What is RetrievalQA?

LangChain ki chain jo retrieved documents ko LLM ke saath combine karke answer generate karti hai.

18. How does RAG reduce hallucination?

LLM ko external documents ka context diya jata hai, isliye answer facts par based hota hai.

#### 19. Difference between Fine-tuning and RAG?

RAG: External documents use karta hai, easy to update knowledge, fast aur cost-effective.

Fine-tuning: Model ke weights change karta hai, retraining ki zarurat hoti hai, zyada expensive.